

Solid-body Impacts on Planets with Thick H/He Envelopes: Penetration, Energy Deposition, and Implications for Super-Puffs

Literature Review Draft — Cursor Preview edition

Notation. Envelope mass fraction

$$f_{\text{env}} \equiv M_{\text{env}}/M_{\text{p}}$$

where M_{env} is envelope mass and M_{p} is planet mass. Impactor mass is M_{imp} .

Citations use AuthorYear keys matching `references.bib` (e.g. Pollack1996). The LaTeX source `lit-review.tex` remains available for journal formatting; **this Markdown file is the primary Preview target.**

Abstract

Close-in low-mass exoplanets span a wide range of bulk densities, from rocky super-Earths to highly inflated “super-puffs” with large hydrogen–helium (H/He) envelope mass fractions. Solid-body impacts can heat, enrich, or erode those envelopes, but the literature is fragmented across giant-impact stripping, formation-stage ablation, thin-atmosphere erosion, and non-impact escape. This review organizes that landscape around a single physical narrative—stopping, energy deposition, and atmospheric fate—and argues that the coupled problem of planetesimal- to lunar-mass solids entering *high- f_{env}* / super-puff envelopes remains open. We do not claim that H/He impacts are unstudied; rather, existing scalings do not yet map penetration depth, envelope-versus-core energy partition, and retention versus loss continuously across super-puff-like envelope fractions.

1. Introduction

Hydrogen–helium envelopes are a central outcome of planet formation and a primary control on the observed radii of low-mass exoplanets (Pollack1996; Lopez2012; Owen2019). For planets that retain primordial gas, f_{env} can range from a thin veneer ($f_{\text{env}} \sim 1\%$) to the extreme values inferred for super-puff planets, whose large radii imply thick, low-density H/He envelopes (Lee2016; Gao2020; Tang2025). How those envelopes respond to late accretion and collisions therefore matters for both formation theory and the interpretation of mass–radius demographics.

The question addressed here is narrower than “do impacts affect H/He atmospheres?” That broader question has already been answered in several adjacent regimes. Giant impacts can strip or thermally inflate H/He envelopes (Inamdar2015; Inamdar2016; Biersteker2019; Denman2020); planetesimals entering forming giant-planet envelopes commonly ablate and deposit mass and energy before reaching the core (Pollack1986; Podolak1988; Valletta2019); and thin atmospheres on terrestrial worlds obey impact-erosion scalings that depend strongly on speed and angle (Schlichting2015; Kegerreis2020a; Kegerreis2020b). What remains unclear—and what motivates the working title of this project—is how *solid bodies* in the planetesimal to lunar-mass range penetrate, deposit energy within, and ultimately modify *high- f_{env}* envelopes, including mature super-puff structures.

This distinction matters because super-puffs are not simply larger mini-Neptunes. Their extended radiative–convective envelopes, low gravities, and long-term mass-loss histories (boil-off, photoevaporation, and related channels) place them in a structural regime where extrapolating thin-atmosphere stripping laws is unsafe

(Owen2016; Ginzburg2018; Misener2021; Tang2025). At the same time, formation-stage ablation calculations were designed for growing gas-giant envelopes, not for the mature high- f_{env} planets of interest here (Pollack1986; Benvenuto2008; Pinhas2016). The novelty of the present framing is therefore not the invention of an empty field, but a newer angle: a systematic mapping of stopping physics, deposition depth, and atmospheric fate across envelope fraction into the super-puff domain.

The key conclusion of this review is correspondingly modest but sharp. The literature already brackets the endpoints—solids can be stopped and ablated in H/He gas, and impacts can remove atmospheres—yet it does not close the intermediate, coupled problem of planetesimal / solid-body entry into high- f_{env} envelopes with depth-dependent deposition and a calibrated loss-versus- f_{env} outcome (Pollack1986; Schlichting2015; Biersteker2019; Chatterjee2018; Tang2025).

Figure 1 (Biersteker & Schlichting 2019, their Fig. 2) shows the thermal-loss process chain after a giant impact: heating, Parker-wind mass loss, and envelope contraction. **Figure 2** (Yang et al. 2026, their Fig. 2) illustrates solid-body penetration and energy deposition in a gaseous envelope (stellar-engulfment hydrodynamics used here as a body-in-gas analogy).

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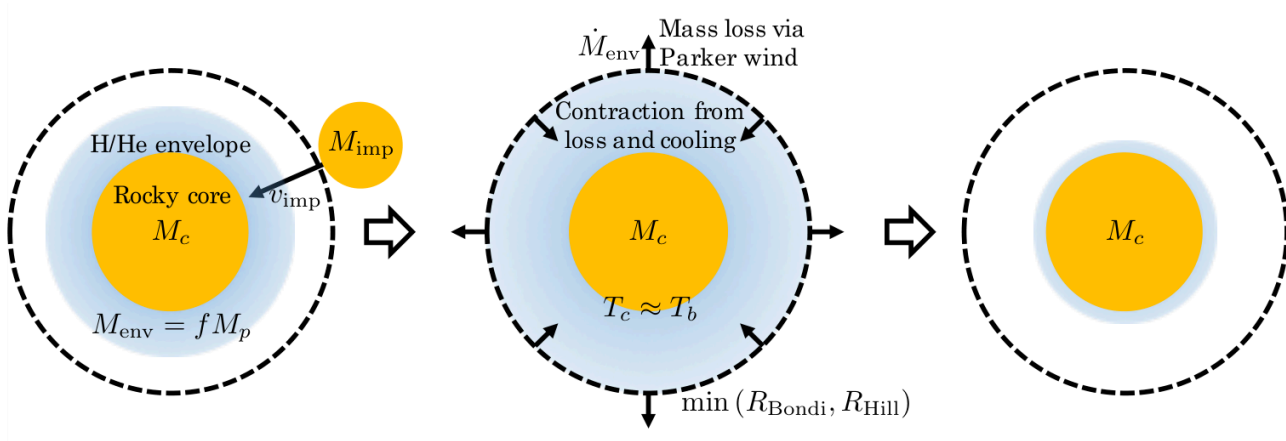


Figure 1. Atmospheric mass-loss pathway after a giant impact. Source: Biersteker & Schlichting (2019), Fig. 2. Educational/review fair use.

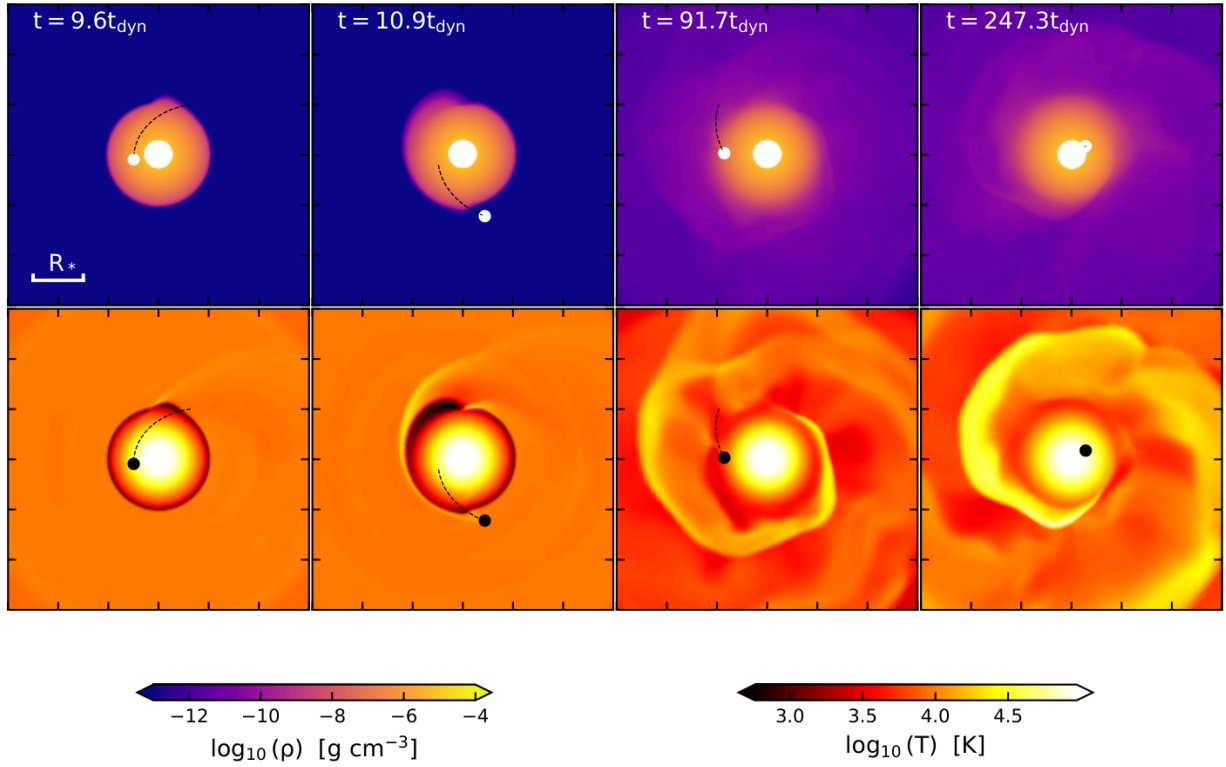


Figure 2. Solid body entering and disturbing a gaseous envelope (density, top; temperature, bottom). Source: Yang et al. (2026), Fig. 2 — cited as a body-in-gas / dynamical-friction analogy, not as a super-puff impact calculation. Educational/review fair use.

2. Distinguishing impact regimes

2.1 Giant impacts versus planetesimal / solid-body entry

Two collision regimes recur throughout the literature and must not be conflated. *Giant impacts* involve embryo-scale impactors whose mass is an appreciable fraction of the target. In that limit, global ground motion, shock launching, and large-scale envelope unbinding dominate the immediate response (Genda2003; Schlichting2015; Inamdar2016; Denman2020). Analytic and numerical work shows that the globally lost atmospheric fraction is often a weaker function of impactor mass and speed than local loss near the impact site, and that complete removal typically requires near-catastrophic conditions (Genda2003; Schlichting2015). For gas-rich planets, SPH and related collision calculations further demonstrate that specific impact energy and projectile-to-target mass ratio control the stripping regime, with grazing encounters much less erosive than head-on ones (Denman2020; Denman2022; Reinhardt2020).

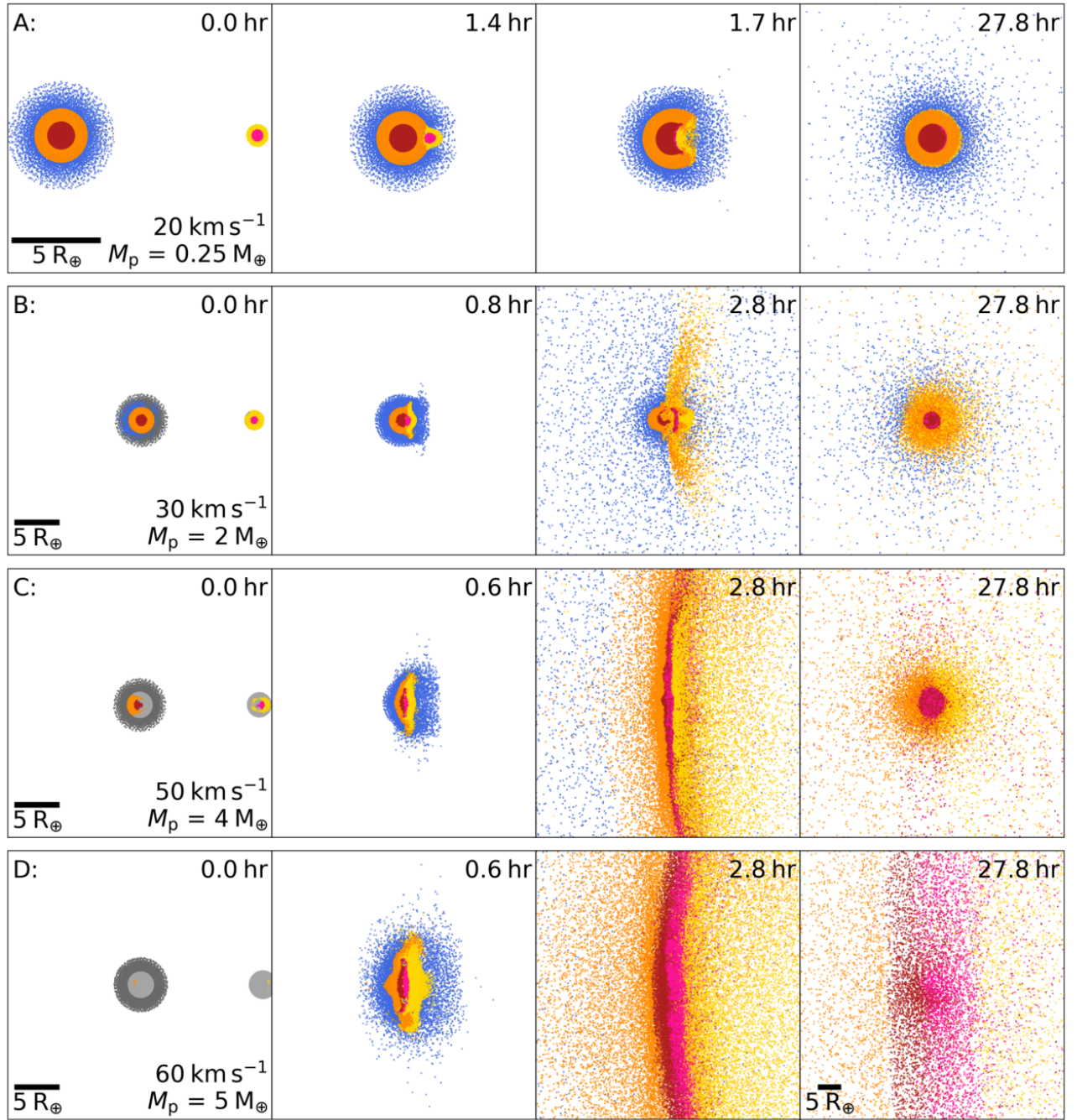


Figure 3. Collision outcomes for atmosphere-bearing targets: merger, atmosphere loss, core/mantle erosion, and supercatastrophic disruption. Source: Denman et al. (2020), Fig. 2. Educational/review fair use.

Planetesimal or, more generally, *solid-body* entry occupies a different physical niche. Here the impactor is small enough that the gaseous envelope is itself the medium through which the body travels. Deceleration is then set by aerodynamic drag, ram-pressure fragmentation, shock heating, and ablation, rather than by a single global shock through the solid planet (Pollack1986; Podolak1988; Korycansky2006). In formation models, this distinction is decisive: many solids never reach the core as intact projectiles, but instead dissolve and deposit their mass and kinetic energy at distributed depths (Pollack1986; Valletta2019; Ormel2021). The working title of this project uses “solid-body” deliberately so that the mass range can extend from hundred-meter planetesimals toward lunar-mass impactors without forcing every case into the giant-impact vocabulary.

2.2 Thin atmospheres versus thick / high- f_{env} envelopes

A second distinction is equally important. Thin-atmosphere erosion studies—often calibrated on terrestrial planets with low atmospheric mass fractions—quantify how much gas is ejected above a tangent plane or unbound by ground motion (Schlichting2015; Kegerreis2020a; Kegerreis2020b). Those scalings are invaluable benchmarks, but they treat the atmosphere as a thin shell to be stripped from a much more massive solid body. In high- f_{env} planets the envelope is a thick, compressible fluid layer whose local column, entropy structure, and heat capacity participate directly in the entry problem (Biersteker2019; Misener2021; Tang2025).

The existence of mature giant-impact calculations for H/He envelopes (Inamdar2015; Biersteker2019; Denman2020) already falsifies any claim that “H/He impacts are unstudied.” The open issue is more specific: whether planetesimal-scale stopping and deposition, when embedded in super-puff-like envelopes, produce enrichment, inflation, delayed escape, or prompt loss as a continuous function of f_{env} .

3. Mapping stopping and breakup in H/He envelopes

3.1 Drag, ablation, and fragmentation

Classic calculations of planetesimal dissolution in forming giant-planet envelopes established the modern hierarchy of entry processes (Pollack1986; Podolak1988). Gas drag decelerates the body; dynamic pressure and thermal ablation remove mass; fragmentation increases the effective cross section and accelerates subsequent deceleration. Small icy bodies tend to dissolve at higher altitude, whereas larger or denser bodies penetrate more deeply, sometimes to the deep envelope or core region (Pollack1986; Pinhas2016; Valletta2019). Ablation can also *increase* capture efficiency by reducing the body’s mass and raising the effective interaction cross section (Benvenuto2008; Inaba2003).

Jovian-entry studies make the same qualitative point with resolved bolide physics: disruption altitude is not identical to the altitude of peak kinetic-energy deposition (Roulston1997; Korycansky2006; Crawford1995). Debris and vapor can continue downward after breakup, so the depth that controls the atmospheric response is the deposition depth, not the first fragmentation height.

3.2 Hydrodynamic drag versus gaseous dynamical friction

For small, non-self-gravitating impactors, aerodynamic (hydrodynamic) drag is the dominant stopping channel in the envelope-entry literature (Pollack1986; Podolak1988; Pinhas2016). Gaseous dynamical friction—the gravitational wake drag formalized for a massive perturber moving through a uniform medium (Ostriker1999; Thun2016)—is conceptually important for larger bodies and for stellar-engulfment analogies (OConnor2023; Yang2026), but direct, like-for-like comparisons of aerodynamic drag versus dynamical friction across the planetesimal-to-lunar-mass range inside planetary H/He envelopes remain sparse. The safest statement supported by the present evidence is therefore asymmetric: drag/ablation is quantified in detail for many formation and entry models, whereas the regime boundary where gravitational drag becomes competitive inside high- f_{env} planetary envelopes is still under-mapped.

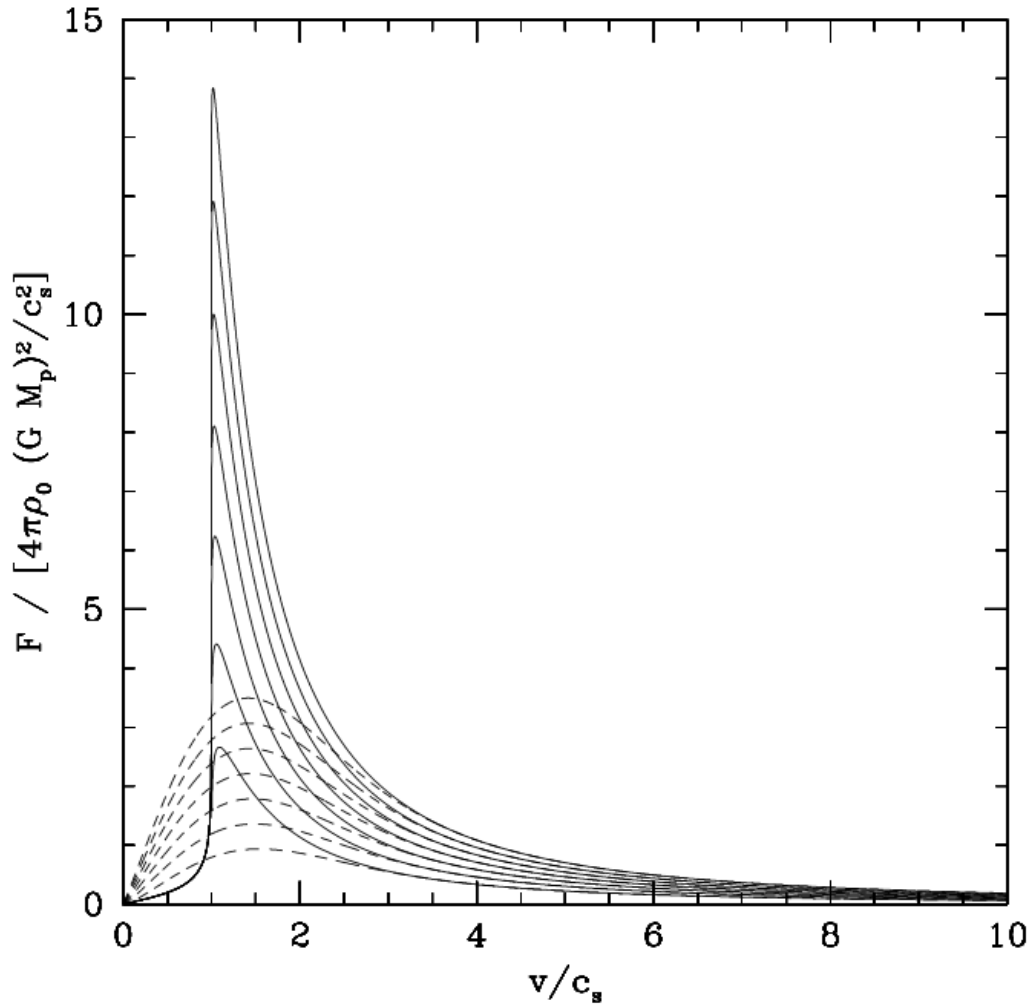


Figure 4. Dynamical friction in a gaseous medium as a function of Mach number v/c_s . Source: Ostriker (1999), Fig. 3. Educational/review fair use.

3.3 Why formation ablation does not automatically close the super-puff problem

Formation-stage ablation is the closest existing analogue for depth-dependent deposition (Pollack1986; Podolak1988; Valletta2019). It is nonetheless incomplete for mature super-puffs. Those calculations target growing proto-giant envelopes with different gravity, entropy, and opacity evolution from the observed low-density exoplanet population (Tang2025; Gao2020). They also emphasize capture and enrichment during accretion, not the post-disk atmospheric fate of a high- f_{env} planet after a later solid-body impact (Chatterjee2018; Biersteker2019). In short, formation ablation supplies the stopping physics, but not the super-puff boundary conditions or the loss-versus- f_{env} map required for the high- f_{env} planetesimal problem.

A practical way to see the mismatch is to ask what quantity one would carry from a formation model into a mature-planet calculation. Capture radius and metallicity of the growing envelope are natural outputs of core-accretion theory (Pollack1996; Valletta2019). For the super-puff problem, however, the needed outputs are different: the radial profile of deposited kinetic energy, the local heating relative to the local envelope binding energy, and the subsequent change in the planet's cooling and escape trajectory (Biersteker2019; Tang2025). Those quantities are rarely tabulated as continuous functions of impactor size and f_{env} for low-gravity, high-entropy envelopes.

4. Connecting energy deposition to atmospheric fate

4.1 Coupling efficiency, not nominal impact energy

Across giant-impact and gas-rich collision studies, atmospheric outcome depends primarily on how much energy and momentum actually couple into the envelope relative to its binding energy and escape conditions (Korycansky1990; Genda2003; Denman2020). Oblique impacts can leave an envelope largely intact even when the nominal impact energy is large; more head-on or faster encounters couple more strongly and eject more gas (Kegerreis2020a; Denman2022). For Uranus/Neptune-class targets, three-dimensional collision models likewise show that grazing encounters can retain or redistribute envelopes rather than fully strip them (Reinhardt2020). The organizing variable is therefore deposited energy (and its depth profile), not the kinetic energy written on the incoming projectile alone.

4.2 Prompt mechanical loss versus delayed thermal escape

For H/He envelopes around low-mass planets, the thermal aftermath of an impact can rival or exceed prompt shock-driven loss (Biersteker2019). Impact heating of the core and envelope can inflate the atmosphere and drive a sustained Parker wind. Crucially, Biersteker2019 identify a heat-capacity transition near envelope fractions of a few percent: when the core dominates the heat budget, a comparatively modest impactor can force strong loss; when the envelope dominates, complete removal instead requires an impactor mass comparable to the envelope mass itself. That result is one of the few quantitative clues that loss physics changes with f_{env} —and it simultaneously warns that extrapolating the low- f_{env} criterion into the super-puff domain is unsafe.

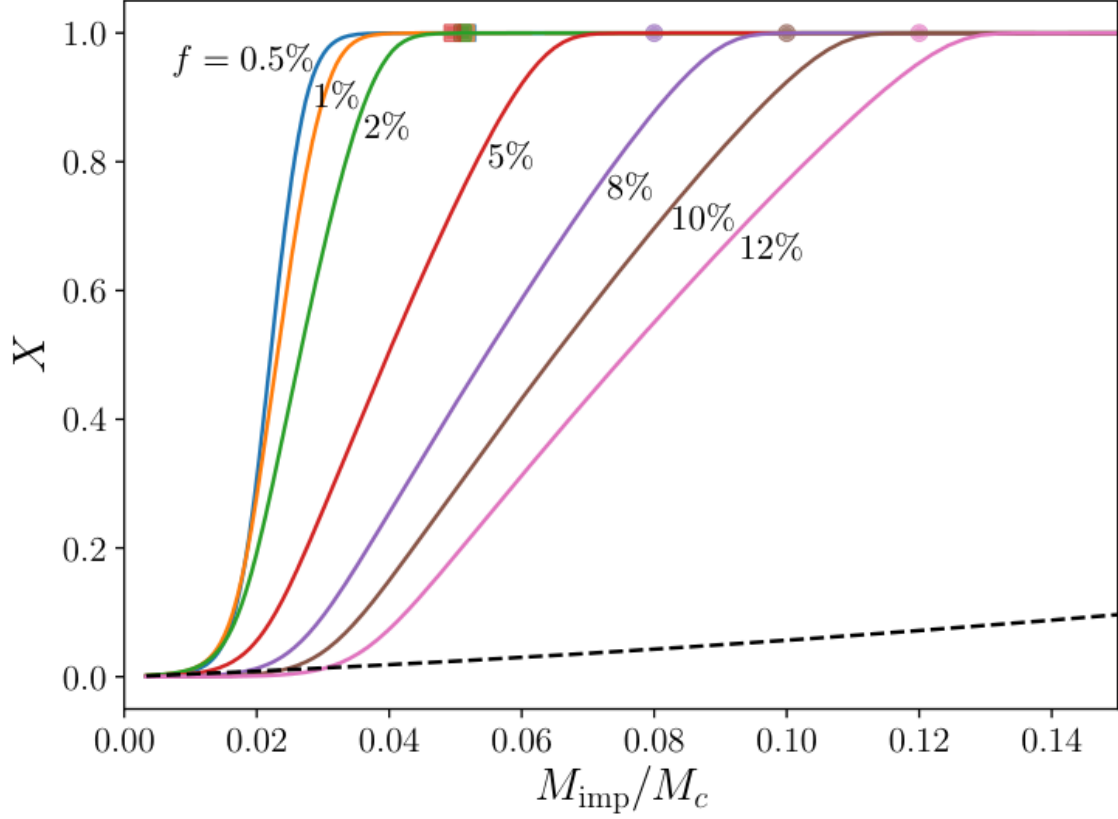


Figure 5. Envelope loss fraction X versus M_{imp}/M_c for different envelope mass fractions f . Higher f requires larger impactors for complete loss; the dashed line marks mechanical-shock estimates. Source: Biersteker & Schlichting (2019), Fig. 5. Educational/review fair use.

4.3 Enrichment as the complementary fate

Impacts and solid accretion do not only remove gas. During giant-planet formation, fragmentation tends to enrich deeper layers while ablation pollutes outer layers, so that much of the incoming heavy material may dissolve in the envelope rather than grow the core (Pollack1996; Valletta2019; Helled2022). Pebble-accretion and polluted-envelope models likewise find vaporization, metallicity gradients, and later rainout (Brouwers2020; Ormel2021; Vazan2023). For sub-Neptunes, planetesimal accretion energy can alter radius and envelope fraction and can enhance subsequent atmospheric loss (Chatterjee2018). These results supply the enrichment and heating branch of the process chain: the same class of solid delivery that strips a weakly bound envelope can, in a more massive envelope, primarily heat and pollute the gas.

Post-impact mixing remains one of the least settled ingredients. Method comparisons suggest that classical SPH can under-mix relative to newer Lagrangian approaches (Deng2019), while improved Jupiter-impact calculations find that heavy elements may re-segregate after transient disturbance rather than persist as a long-lived dilute core

(Sandnes2025; Meier2025). For the super-puff problem, the practical implication is that “where the solid stops” is only the first half of the story; whether deposited metals stay mixed, rain out, or stratify can change opacity, cooling, and long-term radius evolution (Chabrier2007; Leconte2012; Vazan2023).

5. Comparing non-impact loss channels

Any impact-driven narrative for high- f_{env} planets must be embedded in the broader mass-loss landscape. Photoevaporation can carve the evaporation valley and remove light envelopes from close-in planets (Owen2013; Owen2019). Core-powered mass loss provides a complementary channel in which cooling luminosity from the core drives escape, preferentially erasing the lightest envelopes (Ginzburg2018; Misener2021; Owen2024). Boil-off after disk dispersal can remove a large fraction of primordial gas on short timescales (Owen2016). Coupled super-puff evolution models further argue that these planets may be more resilient than simple energy-limited escape predicts, and that boil-off plus photoevaporation already shape much of their history (Tang2025; Gao2020).

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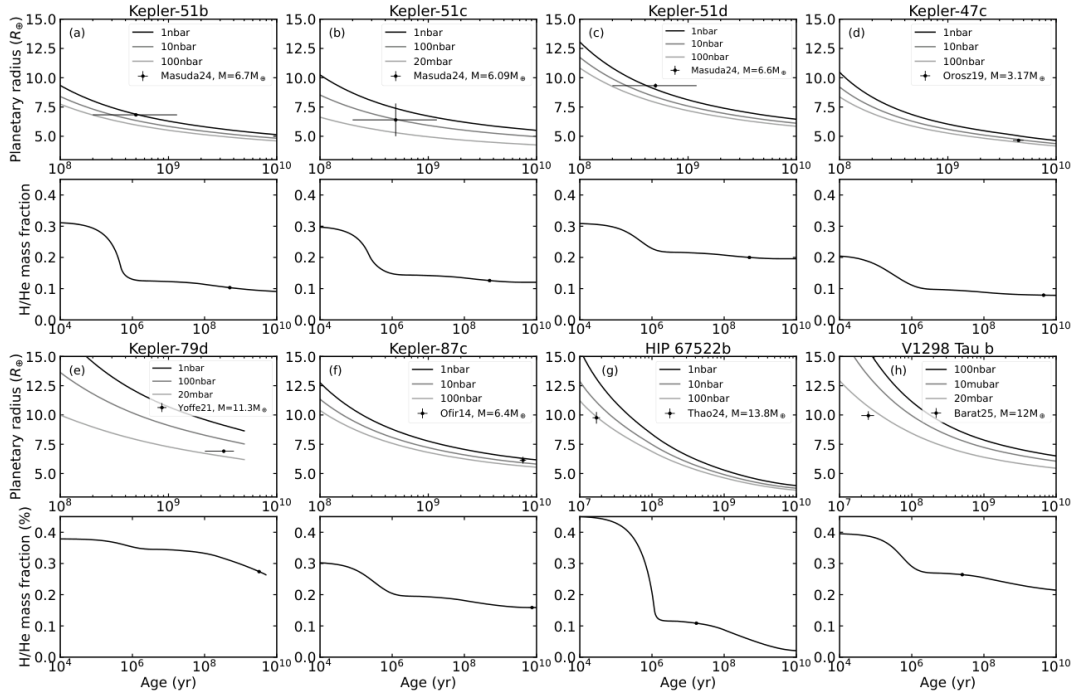


Figure 6. Super-puff / low-density planet evolution under boil-off and photoevaporation: radius and H/He mass fraction versus age. Source: Tang et al. (2025), Fig. 6. Educational/review fair use.

These non-impact channels are not competitors to be ignored; they are the background against which an impact deposition budget must be judged. An impact that deposits heat deep in a high- f_{env} envelope may matter mainly by changing the subsequent Parker-wind or cooling-driven loss rate, rather than by unbinding gas on the dynamical timescale of the collision (Biersteker2019; Chatterjee2018; Tang2025).

The same embedding requirement applies to observational interpretation. Haze opacity can inflate transit radii and “deflate” the inferred need for extremely large envelope fractions (Gao2020), while structure models that include thick radiative atmospheres reduce the H/He mass required to match a given radius (Tang2025). Consequently, even a perfect instantaneous impact calculation is incomplete unless it states whether the deposited heat and

metals are evaluated before or after those evolutionary and atmospheric corrections. For a first toy model, the cleanest path is therefore to fix a hydrostatic envelope at a chosen f_{env} , compute stopping and partition, and only afterward ask how the altered thermal state would evolve under boil-off or photoevaporation (Owen2016; Owen2024; Tang2025).

6. Identifying open questions for high- f_{env} / super-puff planets

The preceding sections support a clear negative conclusion: neighboring literatures illuminate pieces of the problem, but they do not jointly solve planetesimal / solid-body entry into high- f_{env} envelopes with deposition-dependent atmospheric fate. The open questions below are phrased so that an order-of-magnitude toy model—beginning from an adiabatic, hydrostatic H/He envelope—can attack them in sequence.

1. **Stopping column and deposition depth.** For a solid body of radius s and density ρ_s entering a super-puff-like envelope, what overlying column is required for substantial deceleration, and how does deposition depth scale with size, density, and entry speed (Pollack1986; Podolak1988; Korycansky2006)?
2. **Drag versus dynamical friction.** At what impactor mass (or radius) and local Mach number, if any, does gaseous dynamical friction become competitive with aerodynamic drag inside a planetary H/He envelope (Ostriker1999; Thun2016; Yang2026)?
3. **Energy partition.** What fraction of the incoming kinetic energy heats local envelope layers, couples to the deep interior or core, or is carried away with unbound ejecta, as a function of impactor scale and f_{env} (Valletta2019; Biersteker2019; Chatterjee2018)?
4. **Loss versus f_{env} .** How does the few-percent heat-capacity transition of Biersteker2019 connect to the high- f_{env} super-puff regime? Where does the net outcome cross from stripping to enrichment/heating (Biersteker2019; Denman2020; Tang2025)?
5. **Fate of deposited solids.** After entry, do metals remain near the deposition zone, mix globally, or rain out, and how does that choice feed back onto radius and escape (Pollack1986; Ormel2021; Vazan2023)?
6. **Coupling to long-term evolution.** Can an entry/deposition module be embedded in modern super-puff structure and loss calculations so that impact heating is not treated as an isolated impulsive event (Tang2025; Gao2020; Chatterjee2018)?

These questions align directly with the working title: *Solid-body Impacts on Planets with Thick H/He Envelopes: Penetration, Energy Deposition, and Implications for Super-Puffs*. They also respect the literature’s actual boundary conditions. Giant-impact stripping of H/He envelopes is studied; formation ablation is studied; thin-atmosphere erosion is studied. What is missing is the systematic solid-body $\times$ high- f_{env} $\times$ deposition $\times$ fate map.

7. Summary

Solid bodies encountering H/He envelopes usually do not behave like inert projectiles in vacuum. Drag, ablation, and fragmentation determine where they stop; the depth of energy deposition, rather than breakup altitude alone, controls the immediate atmospheric response; and the final envelope fraction depends on both prompt coupling and delayed thermal evolution (Pollack1986; Korycansky2006; Biersteker2019; Tang2025). Giant impacts and thin-atmosphere erosion provide mature scalings at their respective corners of parameter space (Schlichting2015; Kegerreis2020b; Denman2020), while formation-stage models explain distributed enrichment in growing giants

(Valletta2019; Ormel2021). Super-puff planets sit where those corners do not meet. Closing that gap will require a continuous treatment of penetration and energy partition across envelope fraction, followed by an explicit connection to long-term atmospheric fate.

For readers primarily interested in the next calculational step, the review implies a concrete order of operations. First construct a simple adiabatic, hydrostatic H/He envelope over a rocky core and vary f_{env} from thin-atmosphere values into the super-puff range. Second estimate hydrodynamic drag (and, for larger bodies, gaseous dynamical friction) to obtain a stopping depth. Third partition the lost kinetic energy among local envelope heating, deeper interior coupling, and unbound ejecta. Fourth compare the deposited heat to envelope binding energy and to the delayed-escape criteria already calibrated at low f_{env} (Biersteker2019). That sequence does not replace three-dimensional hydrodynamics, but it turns the open gap identified here into a set of falsifiable order-of-magnitude questions rather than a vague call for “more simulations.”

Acknowledgments

This literature synthesis draws on background retrieval reports for orientation, but all scientific claims above are tied to the primary references listed in `references.bib`. Bibliographic descriptors from automated retrieval were cross-checked against ADS/arXiv and local PDF metadata; known mis-associations (for example, citing bombardment-chronology papers as thin-atmosphere erosion frameworks) were removed.

Figure sources (fair use)

File	Source paper	Original figure	Role
<code>figs/fig_Biersteker2019_Fig2_impact_thermal_loss_schematic.png</code>	Biersteker & Schlichting (2019), arXiv:1809.06810	Fig. 2	Impact → thermal loss process
<code>figs/fig_Yang2026_Fig2_envelope_penetration.png</code>	Yang et al. (2026), ApJ 998:118	Fig. 2	Penetration / deposition analogy
<code>figs/fig_Denman2020_Fig2_collision_snapshots.png</code>	Denman et al. (2020), arXiv:2006.01881	Fig. 2	H/He stripping in collisions
<code>figs/fig_Ostriker1999_Fig3_gaseous_dynamical_friction.png</code>	Ostriker (1999), ApJ 513:252	Fig. 3	Gaseous dynamical friction
<code>figs/fig_Biersteker2019_Fig5_envelope_loss_vs_fenv.png</code>	Biersteker & Schlichting (2019)	Fig. 5	Loss vs envelope fraction
<code>figs/fig_Tang2025_Fig6_superpuff_evolution.png</code>	Tang et al. (2025), arXiv:2510.02201	Fig. 6	Super-puff evolution / high f_{env}

Self-drawn schematics formerly in `figs/` have been removed.